MANAKULA VINAYAGAR
INSTITUTE OF TECHNOLOGY
An Autonomous Institution
Affiliated to Pondicherry University, Approved by AICTE, New Delhi,
Accredited by NAAC with 'A' Grade
Kalitheerthalkuppam, Puducherry- 605 107.

FEEDSTOCK QUALITY ASSESSMENT AND DEFECT DETECTION

Department of Artificial Intelligence and Machine Learning

(For Office use only)
Application No.:

DATE:

1. **IP support service required (Tick the appropriate service)**

   a) Patent:

   b) Trademark:

   c) Copyright: ✓

   d) Design:

2. **Point of contact (PoC)**

a) **1st Point of Contact:** Name: Mr. S. Balaji
Mobile No. for contact: 7397363199
WhatsApp No.: 81899 26028
Email: balajisougoumar01@gmail.com

b) 2nd Point of Contact: Name: B. Adharsh Kumar
Mobile No. for contact: 9025134945
WhatsApp No.: 9025134945
Email: adharshkumar2004@gmail.com

c) 3rd Point of Contact: Name: Prasana. B
Mobile No. for contact: 96003 33493
WhatsApp No.: 96003 33493
Email: prasanababu2005@gmail.com

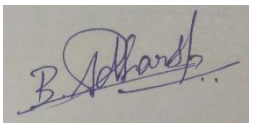
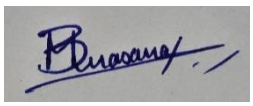
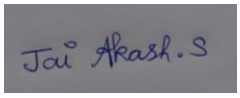
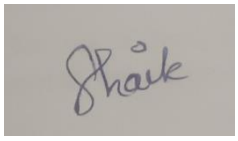
d) 4th Point of Contact: Name: Jai Akash. S
Mobile No. for contact: 6374885639
WhatsApp No.: 6374885639
Email: Sjaiakash8@gmail.com

e) 5th Point of Contact: Name: A. Shaik Roshith
Mobile No. for contact: 9042059574
WhatsApp No.: 9042059574
Email: shaikroshith0000@gmail.com

COPYRIGHT PARTICULARS

CONTACT INFORMATION

1. **Details of the Applicant:**

Name (Official Designation & Address with Phone No)	Permanent Address with Phone No	Age	Experience From To	Citizenship	Mobile No	Signature
Mr. S. Balaji	No.26, Thiyagaraja street, Rajbhavan, Pondicherry, 605001	26	**11 months**	INDIAN	81899 26028	
B. Adharsh Kumar	No.43, 4th Cross, Rainbow Nagar, Puducherry - 605011	21	Pursuing B. Tech	INDIAN	9025134945	
Prasana. B	No.12, First Cross Street, M.G.R Nagar, Moolakulam, Puducherry-605010	20	Pursuing B.Tech	INDIAN	96003 33493	
Jai Akash. S	No:1, Chairman Illam, Earipakkam X-Road, Sooramangalam, Puducherry, 605107.	20	Pursuing B.Tech	INDIAN	6374885639	
A. Shaik Roshith	No,24, Hospital street, Mettupalayam, Puducherry-605009	20	Pursuing B. Tech	INDIAN	9042059574	

2. Communication Details:

Name: Mr. S. Balaji
Address: No.26, Thiyagaraja street, Rajbhavan, Pondicherry-605001
Telephone No: 7397363199
Mobile No: 8189926028
E-mail Id: balajisougoumar01@gmail.com

Name: B. Adharsh Kumar
Address: No.43, 4th Cross, Rainbow Nagar, Puducherry–605011
Telephone No: 9025134945
Mobile No: 9025134945
E-mail Id: adharshkumar2004@gmail.com

Name: Prasana. B
Address: No.12, First Cross Street, M.G.R Nagar, Moolakulam, Puducherry-605010
Telephone No: 96003 33493
Mobile No: 96003 33493
E-mail Id: prasanababu2005@gmail.com

Name: Jai Akash. S
Address: No:1, Chairman Illam, Earipakkam X-Road, Sooramangalam, Puducherry,605107.
Telephone No: 6374885639
Mobile No: 6374885639
E-mail Id: Sjaiakash8@gmail.com

Name: A. Shaik Roshith
Address: No,24, Hospital Street, Mettupalayam, Puducherry-605009
Telephone No: 9042059574
Mobile No: 9042059574
E-mail Id: shaikroshith0000@gmail.com

1. INFORMATION FOR COPYRIGHT FILING

1. **Type of creation (Please tick)** ○ Artistic Works (Poster) ○ Musical Works ○ Literature Works ○ Dramatic Works ○ Books ○ App

   ○ Coding/Hardware ✓

   Annexures Required:

   1. **For APP, Coding/Software**
      Object Code
      Source Code

are required. This has to be given in 2 CDS/2 Pen drives for filing.

2. **For Books, Artistic works (Posters), Literature Works** 2 Hard copies have to be given for filing.
3. **For Musical, Dramatic works**
   The work has to be given in 2 CDS/2 Pendrives for filing.

2. Provide brief description of the functionality/use of your creation.

An AI-Powered Visual Quality Prediction and Defect Detection System is an intelligent industrial solution that uses deep learning and computer vision technology to continuously analyze raw material images and identify defects in real time. Industrial cameras capture images of raw materials such as steel surfaces, automobile components, or machine parts during the manufacturing process. These images are processed using deep learning models to detect minute surface defects such as cracks, scratches, pits, inclusions, and irregularities. The system automatically classifies the material quality and reports defective components to the admin or quality control department through a monitoring dashboard. This reduces manual inspection effort, improves accuracy, and enables faster quality assessment in Industry 4.0 production environments. Additionally, the detected defect data and prediction results are stored for future analysis, helping industries improve manufacturing quality, reduce production losses, and achieve automated smart inspection systems.

3.Whether the work is published or unpublished:

MANAKULA VINAYAGAR
INSTITUTE OF TECHNOLOGY
An Autonomous Institution
Affiliated to Pondicherry University, Approved by AICTE, New Delhi,
Accredited by NAAC with 'A' Grade
Kalitheerthalkuppam, Puducherry- 605 107.
Department of Artificial Intelligence and Machine Learning

FEEDSTOCK QUALITY ASSESSMENT AND DEFECT DETECTION

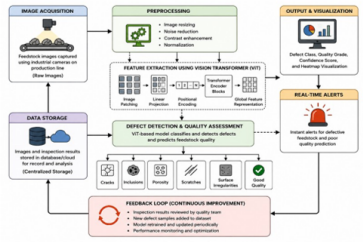
Name of the Students : B. Adharssh Kumar, Prasana. B, Jai Akash. S, A. Shaik Roshith
Dept/Year/Sec. : AI&ML /III/A
Name of the Guide : Mr. S. Balaji,Assistant Professor / AI&ML

ABSTRACT

The proposed Intelligent Vision Transformer-Based Feedstock Quality Assessment and Defect Detection System addresses the growing demand for accurate, automated, and real-time quality inspection in modern Industry 4.0 production environments. This study presents an advanced AI-driven solution that utilizes Vision Transformer (ViT) architecture and computer vision techniques to analyze feedstock surface images for quality evaluation and defect identification with high precision. The system integrates industrial camera-based image acquisition with OpenCV-powered preprocessing and deep learning-based feature extraction to enable fully contactless and non-invasive inspection. Captured images are processed to identify texture inconsistencies, surface irregularities, cracks, inclusions, and other manufacturing defects in real time. The proposed framework ensures consistent quality monitoring across varying lighting conditions and production speeds while minimizing human intervention and manual calibration. By leveraging transformer-based global attention mechanisms, the model achieves superior contextual understanding and defect localization compared to traditional CNN-based inspection systems

EXISTING WORK

Traditional feedstock quality inspection in industrial manufacturing is primarily performed through manual observation, sensor-based analysis, or conventional CNN-based image processing systems. These methods require continuous human supervision and frequent calibration to maintain inspection accuracy across production lines. Manual inspection processes are often time-consuming, inconsistent, and highly dependent on operator expertise, leading to variations in defect identification and material grading.



OBJECTIVE

The objective of the proposed Feedstock Quality Assessment and Defect Detection System is to develop an intelligent Vision Transformer-based platform capable of performing automated, real-time, and contactless inspection of industrial feedstock materials. The system aims to accurately analyze surface characteristics, identify defects, and evaluate material quality using advanced deep learning and computer vision techniques.

INNOVATION

The system uniquely integrates Vision Transformer-based deep learning, intelligent image processing, and automated quality grading mechanisms to deliver accurate defect localization and reliable feedstock quality assessment with minimal human intervention. It enables real-time, contactless inspection by analyzing surface features, texture variations, and visual inconsistencies across production lines.

SDG GOALS



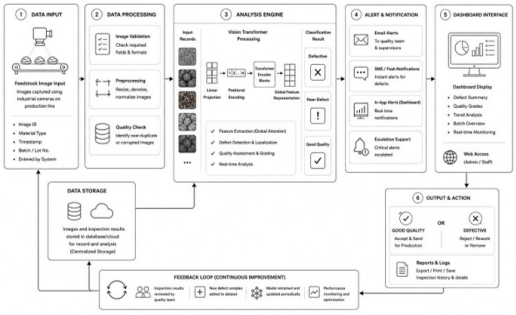
The system integrates Vision Transformers, intelligent image processing, and automated quality grading to deliver accurate defect localization with minimal human intervention.



The proposed solution improves industrial quality monitoring through automated and scalable AI-driven inspection, reducing manual effort while increasing accuracy and consistency.

PROPOSED WORK

The proposed system utilizes a Vision Transformer-based deep learning model integrated with OpenCV and advanced image processing techniques to perform automated, real-time, and contactless feedstock quality assessment and defect detection in Industry 4.0 production environments. The system captures and analyzes feedstock surface images to identify defects, texture inconsistencies, and material irregularities with high precision and reliability. By leveraging transformer-based global feature extraction and intelligent visual analysis, the proposed solution ensures accurate quality grading, reduced human intervention, and improved operational efficiency across automated industrial production lines.



CONCLUSION

The proposed solution provides an intelligent and automated approach for industrial feedstock quality assessment and defect detection through Vision Transformer-based deep learning and real-time image analysis. By enabling contactless and continuous inspection, the system eliminates dependency on manual observation and reduces inconsistencies caused by human error. The integration of AI-driven defect localization, automated quality grading, and real-time monitoring enhances production efficiency, accuracy, and operational reliability across industrial environments. Its scalable architecture supports seamless deployment in Industry 4.0 production systems, enabling faster decision-making, improved quality control, and optimized manufacturing processes. The system is capable of analyzing complex surface textures, identifying minute defects, and maintaining consistent inspection performance under varying lighting conditions and production speeds. Compared to conventional inspection methods, the proposed approach offers higher precision, improved adaptability, reduced operational cost, and enhanced manufacturing consistency. Furthermore, the system establishes a strong foundation for future advancements such as predictive quality analytics, adaptive learning models, edge-based deployment, cloud-integrated monitoring, and fully autonomous smart manufacturing solutions. By combining artificial intelligence, computer vision, and industrial automation technologies, the proposed framework contributes toward the development of smarter, safer, and more efficient next-generation manufacturing ecosystems aligned with the goals of Industry 4.0 and intelligent industrial transformation.

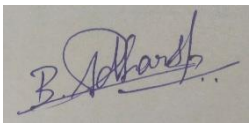
Unpublished

Signature of the Inventor(s)

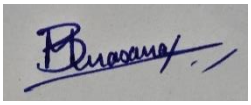
1st Author/Inventor
Name: Mr. S. Balaji
Date
SIGN :



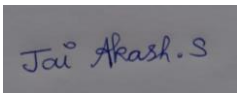
2nd Author/Inventor
Name: B. Adharsh Kumar
Date:
SIGN :



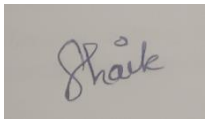
3RD Author/Inventor
Name: Prasana. B
Date:
SIGN:



4th Author/Inventor
Name: Jai Akash. S
Date:
SIGN:



5th Author/Inventor
Name: A. Shaik Roshith
Date:
SIGN:

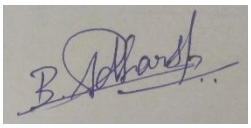
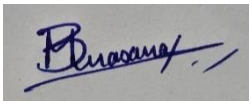
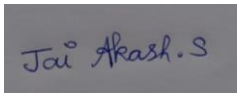


DESIGN PARTICULARS

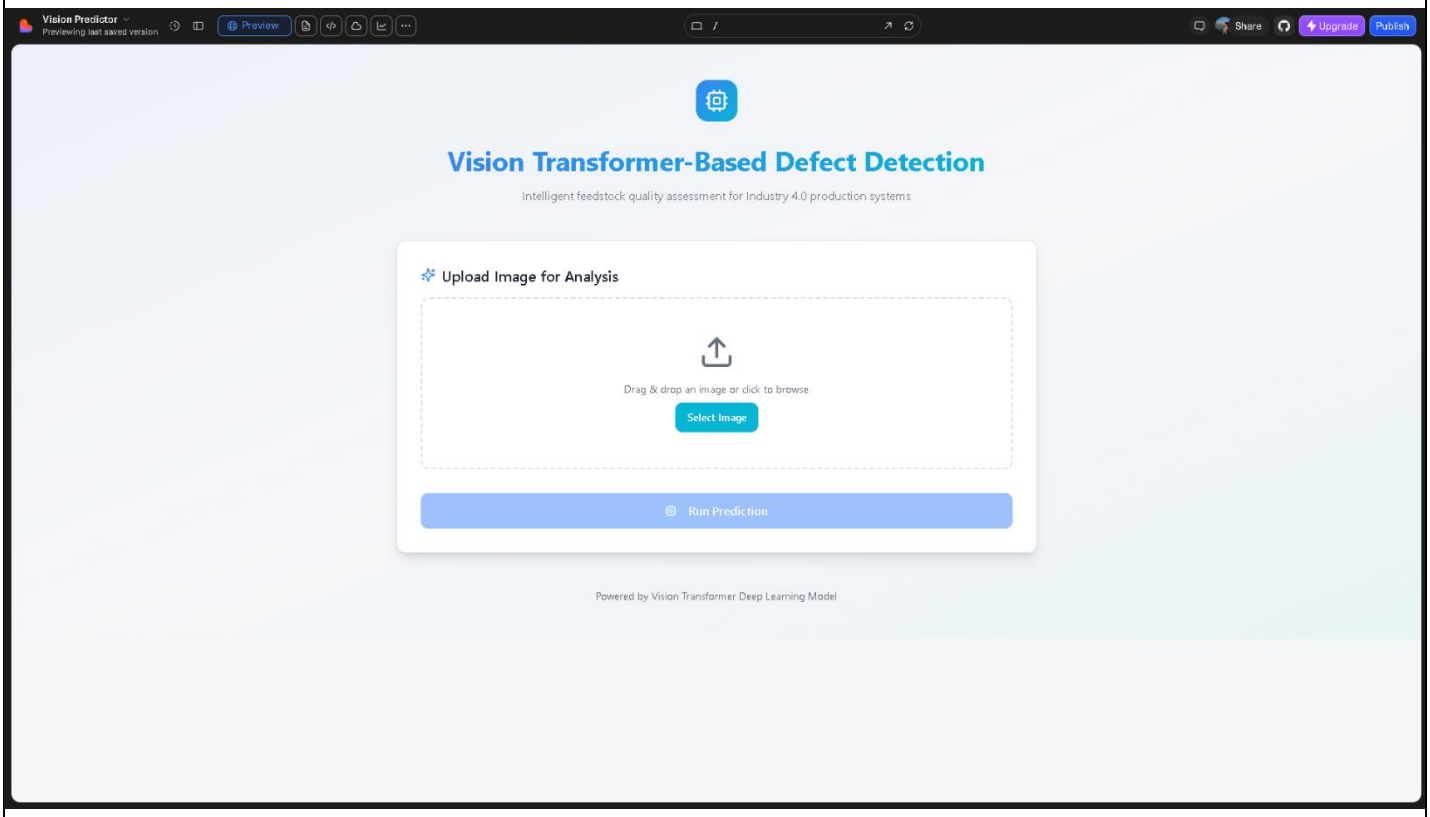
Project Title: FEEDSTOCK QUALITY ASSESSMENT AND DEFECT DETECTION

Inventor Details:

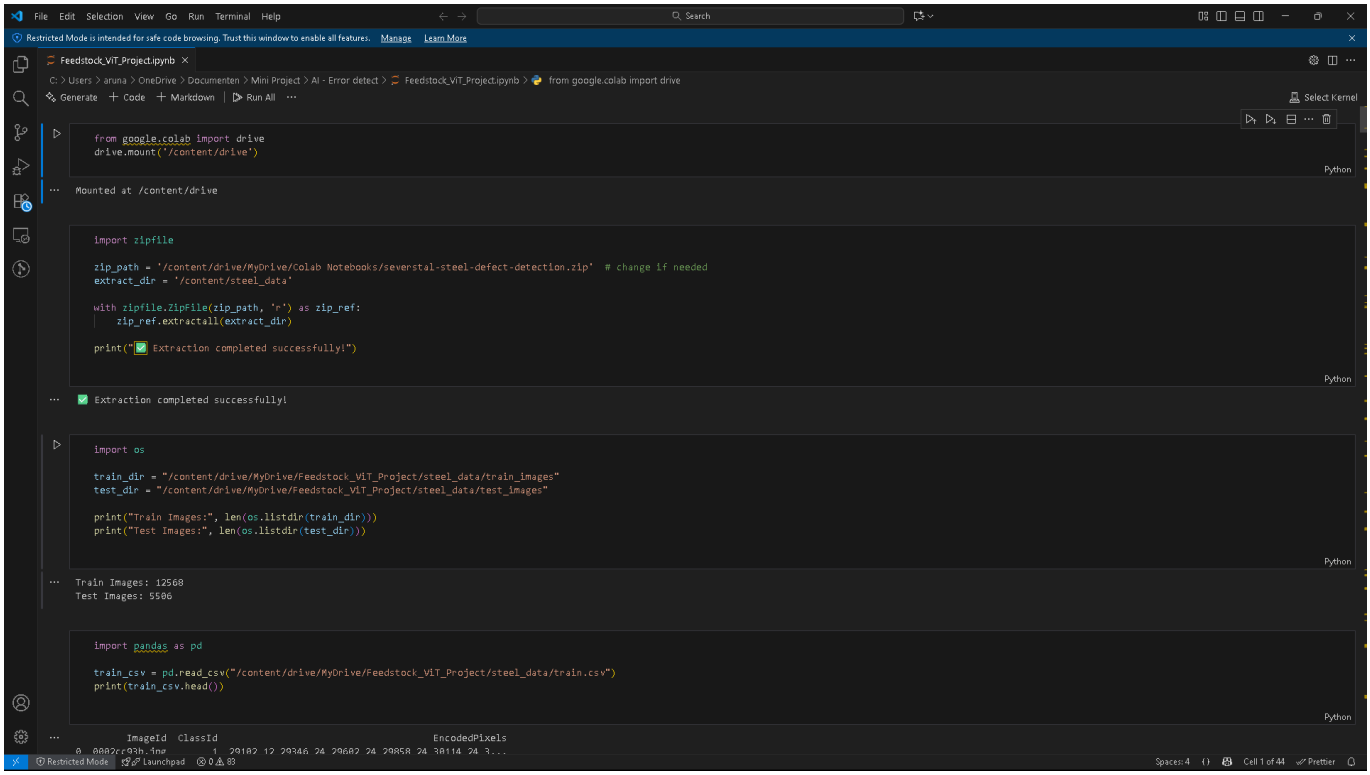
Inventor 1	Name: Mr. S. Balaji	Designation: Assistant Professor / AI&ML
	Phone No: 8189926028	Mail ID: balajisougoumar01@gmail.com
Inventor 2	Name: B. Adharsh Kumar	Designation: Student
	Phone No: 9025134945	Mail ID: adharshkumar2004@gmail.com
Inventor 3	Name: Prasana. B	Designation: Student
	Phone No: 96003 33493	Mail ID: prasanababu2005@gmail.com
Inventor 4	Name: Jai Akash. S	Designation: Student
	Phone No: 6374885639	Mail ID: Sjaiakash8@gmail.com
Inventor 5	Name: A. Shaik Roshith	Designation: Student
	Phone No: 9042059574	Mail ID: shaikroshith0000@gmail.com

Signature of the Inventors			
Inventor 1	Inventor 2	Inventor 3	Inventor 4
			

INFORMATION FOR DESIGN FILING

INPUT


CODE



```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import zipfile

zip_path = '/content/drive/MyDrive/Colab Notebooks/severstal-steel-defect-detection.zip'  # change if needed
extract_dir = '/content/steel_data'

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall(extract_dir)

print("✅ Extraction completed successfully!")
```

```
✅ Extraction completed successfully!
```

```
import os

train_dir = "/content/drive/MyDrive/Feedstock_ViT_Project/steel_data/train_images"
test_dir = "/content/drive/MyDrive/Feedstock_ViT_Project/steel_data/test_images"

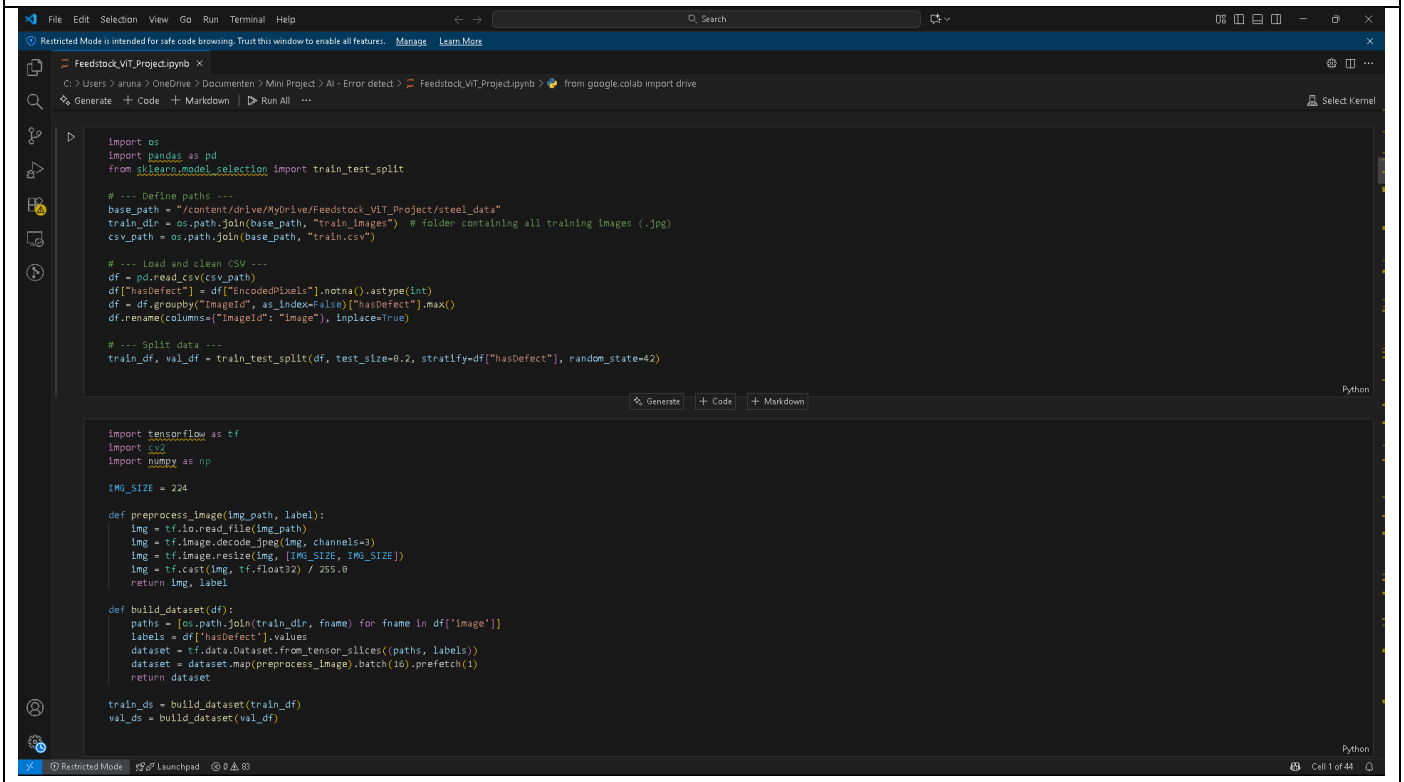
print("Train Images:", len(os.listdir(train_dir)))
print("Test Images:", len(os.listdir(test_dir)))
```

```
Train Images: 12568
Test Images: 5506
```

```
import pandas as pd

train_csv = pd.read_csv("/content/drive/MyDrive/Feedstock_ViT_Project/steel_data/train.csv")
print(train_csv.head())
```

```
           ImageId  ClassId                                       EncodedPixels
0  0002cc93b.jpg        1  29102 12 29346 24 29602 24 29858 24 30114 24 3...
```



```
import os
import pandas as pd
from sklearn.model_selection import train_test_split

# --- Define paths ---
base_path = "/content/drive/MyDrive/Feedstock_ViT_Project/steel_data"
train_dir = os.path.join(base_path, "train_images")  # folder containing all training images (.jpg)
csv_path = os.path.join(base_path, "train.csv")

# --- Load and clean CSV ---
df = pd.read_csv(csv_path)
df["hasDefect"] = df["EncodedPixels"].notna().astype(int)
df = df.groupby("ImageId", as_index=False)["hasDefect"].max()
df.rename(columns={"ImageId": "image"}, inplace=True)

# --- Split data ---
train_df, val_df = train_test_split(df, test_size=0.2, stratify=df["hasDefect"], random_state=42)
```

```
import tensorflow as tf
import cv2
import numpy as np

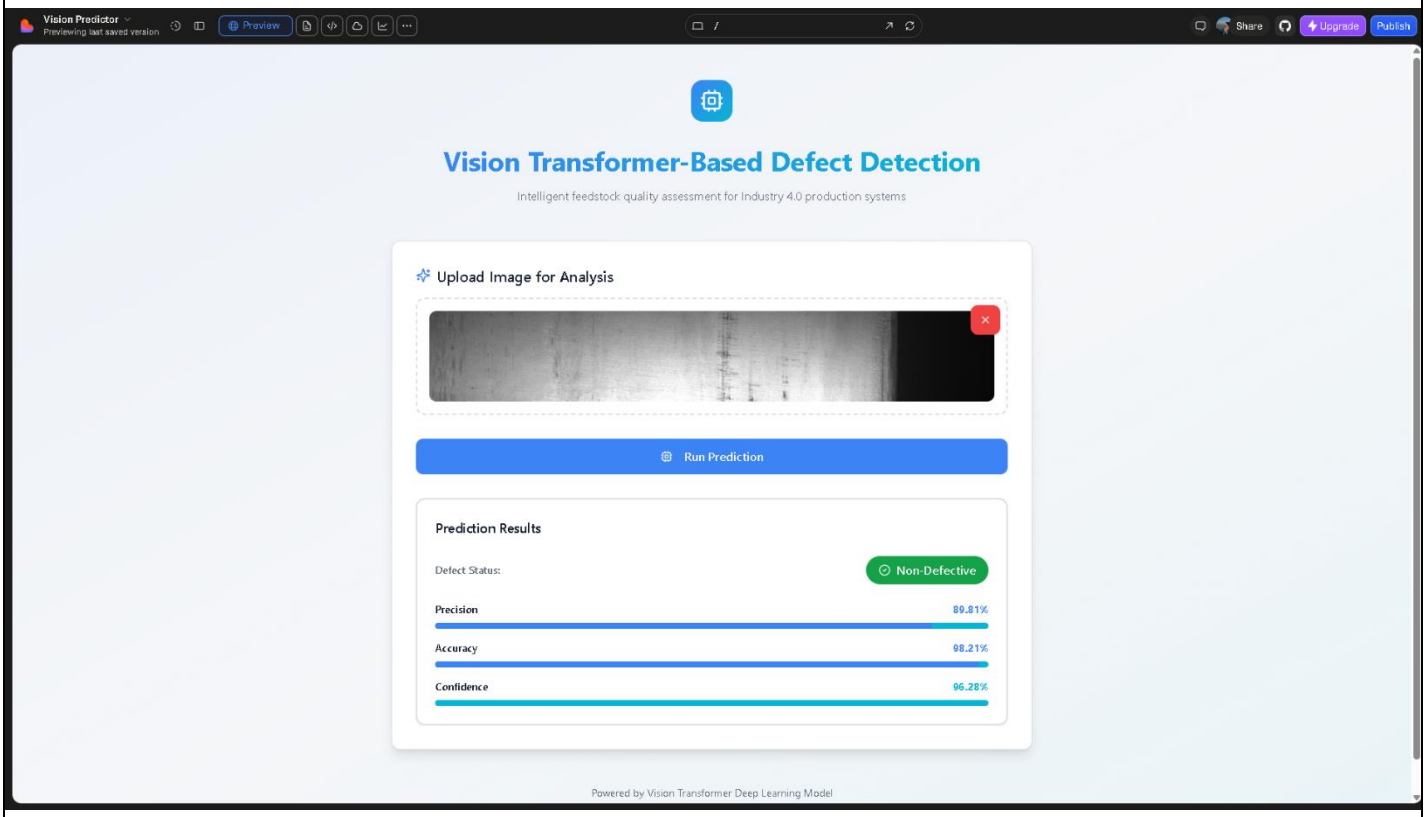
IMG_SIZE = 224

def preprocess_image(img_path, label):
    img = tf.io.read_file(img_path)
    img = tf.image.decode_jpeg(img, channels=3)
    img = tf.image.resize(img, [IMG_SIZE, IMG_SIZE])
    img = tf.cast(img, tf.float32) / 255.0
    return img, label

def build_dataset(df):
    paths = [os.path.join(train_dir, fname) for fname in df['image']]
    labels = df['hasDefect'].values
    dataset = tf.data.Dataset.from_tensor_slices((paths, labels))
    dataset = dataset.map(preprocess_image).batch(16).prefetch(1)
    return dataset

train_ds = build_dataset(train_df)
val_ds = build_dataset(val_df)
```

OUTPUT


Vision Predictor
Previewing last saved version
Preview
Share
Upgrade
Publish
Vision Transformer-Based Defect Detection
Intelligent feedstock quality assessment for Industry 4.0 production systems
Upload Image for Analysis
Run Prediction
Prediction Results
Defect Status:
Non-Defective
Precision
89.81%
Accuracy
98.21%
Confidence
96.28%
Powered by Vision Transformer Deep Learning Model

Abstract: (Minimum 750Words)

The rapid advancement of artificial intelligence and computer vision technologies has greatly improved the efficiency and accuracy of industrial quality inspection systems. An AI-Powered Visual Quality Prediction and Defect Detection System is an intelligent manufacturing solution designed to continuously monitor and analyze raw material images using deep learning and image processing techniques. The system captures high-resolution images of industrial raw materials such as steel surfaces, automobile parts, or machine components and automatically detects defects such as cracks, scratches, dents, inclusions, and surface irregularities. The collected image data is processed using deep learning models to classify materials as good or defective and provide accurate quality assessment results.

This technology has become highly important in modern Industry 4.0 production systems because it supports automated quality inspection, reduces dependency on manual labor, minimizes human error, and improves production efficiency. The system uses industrial cameras and computer vision techniques integrated with deep learning frameworks such as TensorFlow or PyTorch to analyze product quality in real time. The detected defects and prediction results are transmitted to monitoring dashboards or cloud-based systems, allowing administrators and quality control engineers to monitor production quality remotely. The system can also generate automatic alerts whenever defective materials are detected, enabling immediate corrective action and reducing production losses.

Compared to traditional inspection methods, which rely heavily on manual observation and periodic checking, AI-based defect detection systems provide automated, fast, and highly accurate inspection. The architecture of the system generally consists of three major layers: image acquisition layer, processing and analysis layer, and monitoring layer. The image acquisition layer includes industrial cameras responsible for capturing raw material images, the processing layer applies preprocessing and deep learning models for defect identification, and the monitoring layer displays the results through dashboards or reporting systems. Cloud computing and data storage technologies also play an important role in storing large amounts of inspection data securely and making it accessible for future analysis and quality tracking.

One of the major advantages of the AI-Powered Visual Quality Prediction System is real-time defect detection, which allows industries to identify defective materials before they proceed further in the production line. This reduces material wastage, production downtime, and manufacturing costs while improving product reliability and customer satisfaction. The system is especially beneficial in industries such as automobile manufacturing, metal processing, electronics, textile production, and machinery manufacturing where precision and quality control are critical. AI-based inspection systems also improve production data management by automatically maintaining digital inspection records and reducing manual documentation.

However, the system also faces certain challenges and limitations. The accuracy of the deep learning model depends heavily on the quality and size of the training dataset. Variations in lighting conditions, camera angles, and image quality may affect prediction accuracy. In addition, high computational requirements, hardware costs, and integration with existing industrial systems can impact implementation. Despite these challenges, the future of AI-based quality inspection is highly promising due to rapid advancements in deep learning, computer vision, edge computing, and Industry 4.0 technologies. Future systems are expected to include fully automated smart factories, AI-driven predictive maintenance, robotic inspection systems, and real-time industrial analytics.

In conclusion, the AI-Powered Visual Quality Prediction and Defect Detection System is an advanced and efficient industrial solution that enables automated quality inspection, real-time defect detection, and intelligent monitoring. By combining computer vision, deep learning, cloud technologies, and automated reporting systems, the system improves manufacturing quality, reduces operational costs, and enhances industrial productivity. Although challenges related to accuracy, hardware integration, and computational requirements still exist, continuous advancements in AI technologies are making industrial inspection systems smarter, faster, and more reliable. Therefore, AI-based visual quality prediction systems play a major role in the development of modern smart manufacturing and Industry 4.0 production environments.

Application :(Minimum 500 Words)

1. Automated Raw Material Inspection

AI-based quality inspection systems automatically analyze raw material images such as steel sheets, automobile parts, and machine components to detect defects before production begins. This reduces manual inspection time and improves manufacturing accuracy.

2. Surface Defect Detection

Deep learning models identify minute surface defects such as cracks, scratches, dents, pits, and corrosion in industrial materials. The system helps industries maintain high product quality standards.

3. Real-Time Production Line Monitoring

Industrial cameras continuously capture images from conveyor belts and assembly lines. The AI system processes these images in real time and instantly detects defective products during manufacturing.

4. Automobile Component Quality Assessment

The system is used in automobile industries to inspect engine parts, gears, metal sheets, and other components for manufacturing defects. Early detection prevents defective parts from reaching customers.

5. Smart Factory Automation

In Industry 4.0 environments, the system supports fully automated quality control by integrating AI, computer vision, and industrial monitoring systems. This improves production efficiency and reduces human dependency.

6. Defect Reporting and Alert System

Whenever defects are detected, the system automatically generates reports and sends alerts to quality control engineers or administrators. This enables quick corrective action and minimizes production loss.

7. Predictive Maintenance Support

Continuous monitoring of machine parts and raw materials helps identify wear and damage at early stages. This supports predictive maintenance and prevents unexpected machine failures.

8. Industrial Data Analysis and Record Management

The system stores inspection results, defect images, and quality reports in databases or cloud platforms. Industries can use this data for quality analysis, auditing, and production optimization.

9. Metal and Steel Industry Inspection

AI-based defect detection systems are widely used in steel and metal industries to inspect surface quality, detect rolling defects, and ensure compliance with industrial standards.

10. Electronics and Precision Manufacturing

The system can inspect electronic components, circuit boards, and precision-manufactured products where even tiny defects can affect performance and reliability.

Advantages :(Minimum 250 Words)

1. Real-Time Defect Detection

The system continuously monitors raw material and product images to detect defects instantly during the manufacturing process.

2. Automated Quality Inspection

AI-based inspection reduces the need for manual quality checking and improves consistency in defect identification.

3. Early Detection of Manufacturing Defects

Minute defects such as cracks, scratches, dents, and surface irregularities are identified at early stages, preventing defective products from entering production.

4. Reduced Production Costs

Early defect identification minimizes material wastage, product rejection, rework, and maintenance costs in industries.

5. Improved Product Quality

The system ensures high-quality manufacturing standards by accurately classifying defective and non-defective materials.

6. Increased Industrial Productivity

Automated inspection speeds up the quality control process and improves overall production efficiency.

7. Automatic Data Collection and Storage

Inspection results, defect images, and quality reports are stored automatically in databases or cloud systems for future analysis.

8. 24/7 Continuous Monitoring

The system operates continuously without interruption, enabling round-the-clock industrial quality inspection.

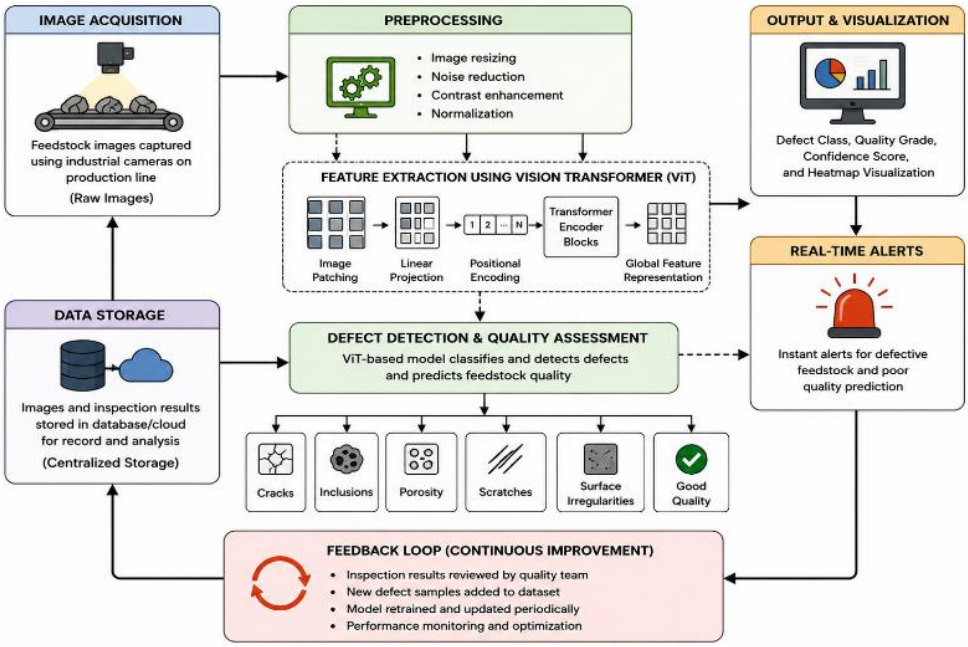
9. Easy Access to Inspection Reports

Quality engineers and administrators can easily access defect reports, production history, and inspection records through dashboards or monitoring systems.

10. Supports Smart Manufacturing and Industry 4.0

The project contributes to the development of intelligent factories by integrating artificial intelligence, computer vision, and automated industrial monitoring systems.

Existing System Sample Drawing (not mandatory)



Proposed system Sample Diagram

